Survive the Field: Why the BBP Machine Has Its Form

Rational wheels, rigid lanes, layers as atoms, nesting depth, and the same Mold in a compiler

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Abstract

This sequel to *Nest, Run, Fold* [1] asks why the BBP digit machine has the Form found there, and whether that Form belongs to π or to any machine that transforms without forgetting. The guiding picture is a runner crossing a field: nothing is accumulated, the starting point fixes how much field lies ahead, and a process is complex to the extent that it survives the whole field. Five answers come from the machine itself. The fold wheels $s = 1/(1 - z)$, $|z| = 1$, that close on a rational number are exactly the crystallographic rotations; only the quarter and third turns both turn and shrink, and base 16 is $(1 - i)^8$. By Baker's theorem the eight base-16 lanes satisfy exactly one rational relation, so the Adamchik–Wagon line holds every π formula on them, while the twelve base-729 lanes of the 30° wheel satisfy two. The four lanes of a layer are copies of one motion that agree on about $2 \log_{16} k$ digits, and a layer is heard only through their difference. Each vanishing moment of the weights adds log k digits of silence, and as many can vanish as there are relations: one for BBP, two for the ternary machine, which has a unique doubly crested formula for $\pi\sqrt{3}$. An involution in a loop silences exactly the harmonics it reflects. The same Mold then appears in a compiler. Each SHA-256 round and its message schedule are Molds, so for every message the 64 rounds permute the state and keep its past with no stack. A one-bit change enters at the round of its word, crosses a four-stage delay line, binds every state bit to every bit of its word five rounds later, and cannot vanish while both runs execute the same program. In both machines distinctness is lost in one place only: where the output drops the half of the Mold that was retained.

Keywords: BBP formula; Baker's theorem; crystallographic restriction; cyclotomic fields; reversible dynamics; Feistel networks; SHA-256; diffusion. *MSC 2020:* 11Y60, 11J86, 11R18, 11A63, 94A60.

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1 Introduction: the field

Nest, Run, Fold [1] read the formula of Bailey, Borwein and Plouffe [6],

$$\pi = \Sigma_{k\geq 0}\, 16^{-k}\,(\,4/(8k+1) - 2/(8k+4) - 1/(8k+5) - 1/(8k+6)\,), \tag{1}$$

as a machine. Every term is a phase $\{16^{n-k}/(8k+j)\}$ on one circle, present from step 0 and moved by the single map $x \mapsto \{16x\}$. A *lane* is a pair (k, j) with modulus 8k + j and weight $c_j \in \{4, -2, -1, -1\}$; a *layer* is the four lanes with the same k, and its value is $V_k = \Sigma_j\, c_j/(8k+j)$. One Form organized everything found there: compile first; nest until nesting can no longer hold a term, then run; encapsulate; invert and fold; retain the prior form so that every step can be undone, the Mold $G(a, b) = (b, F(b) \star a)$ [1, §7]; and read the difference a step produces. That paper shows the Form operating. This one asks why the machine has it, and whether it belongs to π or to any machine that transforms without forgetting.

The guiding picture is a runner crossing a field. The runner accumulates nothing. The starting point fixes how much field lies ahead, the field fixes the layers that must be crossed, and once the far zone is reached it closes on its own. A process is complex, in this picture, not because it accumulates mass but because it survives the whole field. In the BBP machine the runner is a layer, and its field is the depth it nests before its value must run. In a compiler the runner is a change in the source, and its field is the run of the compiled program.

Sections 2 and 3 find the simplest instructions a rational digit machine can carry, the rational wheels, and count what their lanes can hold. Sections 4–6 follow what copies of one motion are forced to do: agree, split, nest and lean. Section 7 finds the same Mold in SHA-256 and maps the field of a message word. Notation follows [1]: $L_j = \Sigma_{k\geq 0}\, 16^{-k}/(8k+j)$ for j = 1, …, 8; the Adamchik–Wagon line [2]

$$\pi = \Sigma_{k\geq 0}\, 16^{-k}\, \Sigma_{j=1}^{8}\, c_j(r)/(8k+j), \quad c(r) = b - rR, \tag{2}$$

with b = (4, 0, 0, −2, −1, −1, 0, 0) and the zero relation R = (−8, 8, 4, 8, 2, 2, −1, 0), so that BBP is r = 0; the crossing $q_{k,j}(n) = \lfloor 16x_{k,j}(n) \rfloor$ of a lane at step n; and the layer's signed crossing $H_k(m) = \Sigma_j\, c_j\, q_{k,j}(k + m)$ at age m. Results of [1] are cited by their numbers there. Table 1 separates what is proved, what is measured and how far, and what is open; every number comes from a script listed in the Appendix.

Table 1. Results ledger.

Result	Status	Origin
Thm. 1. The rational fold wheels are the crystallographic rotations; only the orders 2, 3, 4 shrink; $16 = (1 - i)^8$ and $729 = (1 - \omega)^{12}$.	Proved	Niven [19]; crystallographic restriction; arrangement new
Thm. 2. Exactly one rational relation among the eight base-16 lanes, exactly two among the twelve base-729 lanes.	Proved (Baker [8]) with an exact finite computation	Relations: [3, 5]; completeness: new
Prop. 3. A layer is four copies that agree on about $2 \log_{16} k$ digits; its signal starts exactly where they split.	Proved; 2,700 layers up to $k \approx 2 \times 10^7$	New
Prop. 4. Silent depth $(\nu + 1) \log_b(Nk)$ for ν vanishing moments; $\nu \leq 1$ on the binary lanes, $\nu = 2$ once on the ternary lanes.	Proved; Table 2	New

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Result	Status	Origin
Prop. 5, Obs. 6. Which harmonics a chord lets through; which loops carry the drift.	Proved; numerical, $k \le 128{,}000$	New
Thm. 7. Every SHA-256 round and its schedule are Molds; for each block the 64 rounds permute the state.	Proved; run backwards exactly	Structure: [13, 18]; reading: ours
Prop. 8. In both machines distinctness is lost only where the output drops the retained half of a Mold.	Proved	New in this form
Prop. 9. A change in word j: first contact at round j, delay line to j + 3, no vanishing before round τ_j.	Proved; 10,752 one-bit changes	New
Obs. 10. Every state bit binds to every bit of W_j at round j + 5; values saturate one round later; the value front can lag.	Structure computed exactly; values measured; lag shown by an explicit block	New
Whether every fold system spans as many dimensions as its fold logarithms; normality and the traveler [1, 7].	Open	—

2 Rational wheels: the first instructions

A compiler starts from its simplest instructions. For a digit machine on a circle these are the turns that close on a rational number. The wheel behind BBP is $s = (1 + i)/2$, and [1, Cor. 7] rests on its fold $s + \bar{s} = 1$. Every point of the line $\mathrm{Re}\, s = ½$ has this fold, and the line is the image of the unit circle under $z \mapsto 1/(1 - z)$, since $|1/s - 1| = 1$ exactly when $|1 - s| = |s|$. A fold wheel $s = 1/(1 - z)$, $z = e^{i\theta} \neq 1$, turns by $t = \arg s = \pi/2 - \theta/2$ and shrinks by $|s| = 1/|1 - z|$ at each step, and

$$\Sigma_{n\ge1}\, s^n/n = -\log(1 - s) = -\log \bar{s} = \ln(1/|s|) + it, \tag{3}$$

so its two shadows are a logarithm and an angle. A digit machine needs more: some power of s must be a rational number, the base at which the wheel closes and a layer ends.

Theorem 1 (rational fold wheels). *Let $s = 1/(1 - z)$ with $|z| = 1$ and $z \neq 1$.*

(a) Some power s^N, $N \ge 1$, is rational if and only if z is a root of unity of order 2, 3, 4 or 6, that is, if and only if $2\,\mathrm{Re}\, z$ is an integer. These are exactly the nontrivial orders of rotations that preserve a lattice in the plane (the crystallographic restriction).

(b) The wheel shrinks, $|s| < 1$, so that its lanes converge geometrically with base $|s|^{-N} > 1$, for the orders 2, 3 and 4, and not for order 6, where $|s| = 1$ and the base is 1. Equivalently, $1 - z$ is a prime for the prime-power orders, namely 2, $1 - i$ over 2, and $1 - \omega$ over 3 ($\omega = e^{2\pi i/3}$), and a unit for order 6.

(c) The shrinking wheels are $s = ½$ (no turn: log 2 in base 2), $s = 1/(1 - i) = (1 + i)/2$ (45°, closing at $s^8 = (1 - i)^{-8} = 1/16$: π and log 2 in base 16) and $s = 1/(1 - \omega) = e^{i\pi/6}/\sqrt{3}$ (30°, closing at $s^{12} = (1 - \omega)^{-12} = 1/729$: $\pi\sqrt{3}$ and log 3 in base 729). Their half-turns $s^4 = -1/4$ and $s^6 = -1/27$ give alternating forms in bases −4 and −27.

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(d) One turn sums to $\Sigma_{j=1}^{N} s^j = (1 - |s|^N)\, e^{2it}$, which is purely imaginary only for the 45° wheel: it is the only shrinking rational wheel whose fold shadow lands on its crest by itself.

Proof. (a) If $s^N = q \in \mathbb{Q}$, then $N \arg s \in \pi\mathbb{Z}$, so z is a root of unity, and $\beta = |1 - z|^2 = 2 - z - z^{-1}$ satisfies $\beta^N = q^{-2}$. β is an algebraic integer whose conjugates $2 - z' - z'^{-1}$ are positive reals with the same N-th power, hence all equal to β; so $\beta \in \mathbb{Z}$, and $0 < \beta \le 4$ leaves $\beta \in \{1, 2, 3, 4\}$, that is, $\mathrm{Re}\, z \in \{\tfrac{1}{2}, 0, -\tfrac{1}{2}, -1\}$. Conversely these give $s = e^{i\pi/3}$, $(1 + i)/2$, $e^{i\pi/6}/\sqrt{3}$ and $\tfrac{1}{2}$, whose powers s^6, s^8, s^{12} and s are rational. This is Niven's argument [19, Ch. 3]. A rotation that preserves a lattice has an integer matrix in a lattice basis, so its trace $2\cos\theta = 2 - \beta$ is an integer: the same condition. (b) $|s| = \beta^{-1/2} < 1$ if and only if $\beta \ge 2$; $2 = i(1 - i)^2$, $3 = -\omega^2(1 - \omega)^2$, and $1 - e^{i\pi/3} = e^{-i\pi/3}$. In general $1 - \zeta$ is a unit when the order of ζ is not a prime power [21, Prop. 2.8]. (c) $t = \pi/2 - \theta/2$ is 0, $\pi/4$ or $\pi/6$; group $n = Nk + j$ as in [1, Thm. 6]. The half-turn powers are $s^{N/2} = -|s|^{N/2}$. (d) The turn sum is $s(1 - s^N)/(1 - s) = (1 - |s|^N)\, s/\bar{s}$, and $s/\bar{s} = e^{2it}$. ■

Base 16 is therefore not a choice. Within the fold, a rational machine can turn only by the crystallographic angles; only the prime-power turns shrink, so that something can nest; and of those only the quarter turn carries an angle in a base that is a power of 2. Its step is the inverse of the prime $1 - i$, and one machine step, multiplication by $16 = (1 - i)^8$, is eight such primes [1, Thm. 6e]. The 30° wheel is the same construction over the prime $1 - \omega$, and its two shadows are exactly Adegoke's ternary formulas for $\log 3$ and $\pi\sqrt{3}$ in base 729 [3, (2.11), (2.13)]. The hexagonal turn closes ($s^6 = 1$) but never shrinks, so nothing in it can nest: it is a loop without layers (Fig. 1A). Off the fold line other BBP-type formulas exist, $\pi\sqrt{3}$ in base 64 among them [5], and so do formulas in irrational bases such as the golden ratio [15]; the theorem concerns the fold alone, the Form's $\tfrac{1}{2} + \tfrac{1}{2} = 1$.

3 Rigid and flexible lanes

[1, Thm. 5] singles out BBP on the Adamchik–Wagon line, but that settles BBP among all formulas on its eight lanes only if the line is all there is: any further relation among $L_1, \ldots, L_8$ would give further π formulas on the same lanes. Integer-relation searches find only R, and several authors generate BBP-type formulas systematically [9, 15], but a search cannot show that nothing is missing. The next theorem does, and the same argument applied to the 30° wheel gives a different answer. Write $L'_j = \Sigma_{k\ge 0}\, 729^{-k}/(12k+j)$ for $j = 1, \ldots, 12$, and $\zeta_N = e^{2\pi i/N}$.

Theorem 2 (rigid and flexible lanes). *(a) The L_j are linear combinations, with coefficients in $\mathbb{Q}(\zeta_8)$, of seven logarithms: $i\pi$, $\ln 2$, $\ln(1 + \sqrt{2})$, $\log(\sqrt{2} - i)$, $\log(\sqrt{2} + i)$, $\log(2 + i)$ and $\log(2 - i)$. The L'_j are combinations, with coefficients in $\mathbb{Q}(\zeta_{12})$, of ten: $i\pi$, $\ln 2$, $\ln 3$, $\ln(2 + \sqrt{3})$ and $\log(\sqrt{3} - \zeta_{12}^{a})$ for $a = 2, 4, 5, 7, 8, 10$.*

(b) The rational relations $\Sigma_j q_j L_j = 0$ are exactly the multiples of R. The rational relations $\Sigma_j q_j L'_j = 0$ form a plane, spanned by $A = (0, 81, -162, 0, 27, 36, 0, 9, 6, 4, -1, 0)$ and $B = (243, -405, 0, -81, -27, -72, -9, -9, 0, -5, 1, 0)$.

(c) Every formula $\pi = \Sigma_j c_j L_j$ lies on the line (2), and BBP is the only one with $\Sigma_j c_j = 0$. The formulas $\pi\sqrt{3} = \Sigma_j c_j L'_j$ form a plane.

Proof. Put $p = 2$, $N = 8$ or $p = 3$, $N = 12$, and $\zeta = \zeta_N$. Filtering $-\log(1 - x) = \Sigma\, x^n/n$ by $n \bmod N$ at $x = p^{-1/2}$ gives $L_j = -(p^{j/2}/N)\, \Sigma_{a=0}^{N-1}\, \zeta^{-aj} \log(1 - \zeta^a/\sqrt{p})$, and $1 - \zeta^a/\sqrt{p} = (\sqrt{p} - \zeta^a)/\sqrt{p}$. For $p = 2$, $\sqrt{2} - 1 = (1 + \sqrt{2})^{-1}$, $\sqrt{2} - \zeta = \zeta^{-1}$, $\sqrt{2} - \zeta^3 = \zeta^{-1}(2 + i)$, $\sqrt{2} - \zeta^5 = \zeta(2 - i)$, and $\sqrt{2} \mp i$ are the two primes of $\mathbb{Q}(\zeta_8)$ over 3. For $p = 3$, $(\sqrt{3} \mp 1)^2 = 2(2 \mp \sqrt{3})$, $\sqrt{3} \mp i = 2\zeta^{\mp 1}$, $\sqrt{3} - \zeta^{\pm 1} = \zeta^{\mp 1}$, $\sqrt{3} - \zeta^{\pm 5}$ are the two primes over 7, and the remaining four are the

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four primes over 13. This gives (a), each logarithm on its principal branch. The logarithms in (a) are those of −1, of a unit of infinite order and of numbers that generate powers of distinct prime ideals, which are multiplicatively independent modulo roots of unity; so they are linearly independent over $\mathbb{Q}$ and, by Baker's theorem [8, Thm. 2.1], over the algebraic numbers. A rational relation among the lanes is therefore exactly a rational solution of the linear system that makes every coefficient in (a) vanish. Written in the basis 1, ζ, ζ^2, ζ^3 of each field, the two systems have rank 7 and 10, and R, respectively A and B, span their solutions. (c) follows from (b) and [1, Thm. 5]. ■

R is known [5], and A and B span the two zero relations that Adegoke gives in base 729 [3]; what the theorem adds is that there are no others, and we have not seen this stated. The counting behind it is the point. In both systems the lanes span exactly as many dimensions as there are logarithms behind their fold points: seven for eight lanes, ten for twelve. A lane is a slot, and what the slots can hold is fixed by the primes into which the points $\sqrt{p} - \zeta^a$ factor, 3 and 5 for the binary wheel and 7 and 13 for the ternary one, together with the angle, the base and one unit. The binary machine has one slot more than it has logarithms, so there is exactly one way to trade its lanes against each other: it is rigid. The ternary machine has two ways, and its formulas form a plane.

On that plane, $c = u + aA + bB$ with $u = (243, 243, 162, 81, 27, 0, -9, -9, -6, -3, -1, 0)/81$ the 30° wheel, the weight sum is $728/81 - 364b$, so the zero-sum formulas form a line. The sparsest formulas use six lanes, and both are alternating series read into base 729: the Madhava–Sharp series $\pi\sqrt{3} = 6\,\Sigma_{n\ge0}\,(-1/3)^n/(2n+1)$, and $\pi\sqrt{3} = \Sigma_{k\ge0}\,(-1/27)^k\,(6/(6k+2) + 6/(6k+3) + 2/(6k+4))$. The sparsest zero-sum formula uses seven,

$$9\pi\sqrt{3} = \Sigma_{k\ge0}\,729^{-k}\,(\,81/(12k+1) - 54/(12k+2) - 9/(12k+4) - 12/(12k+6)$$
$$- 3/(12k+7) - 2/(12k+8) - 1/(12k+10)\,), \qquad (4)$$

the ternary counterpart of BBP: the crest member that uses the fewest lanes.

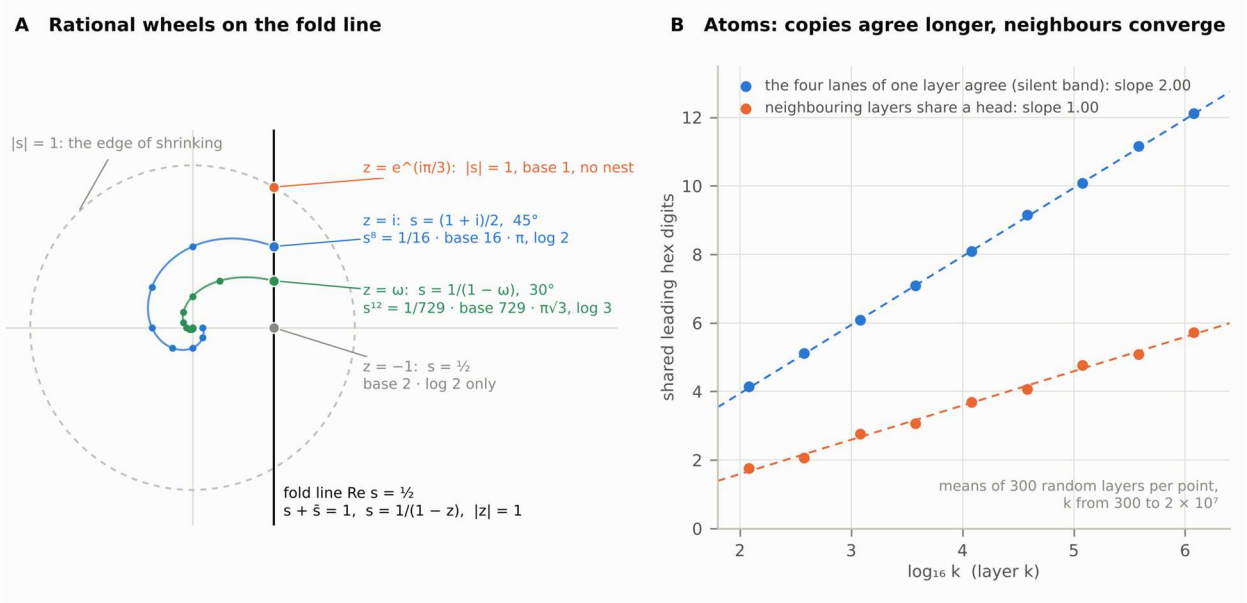


Figure 1. Why this wheel, and why atoms. (A) The fold line Re s = ½ is the image of the unit circle under $z \mapsto 1/(1 - z)$. The points whose wheels close on a rational number are the crystallographic rotations (Theorem 1): z = −1 (s = ½, no turn), z = i (45°, $s^8 = 1/16$), z = ω (30°, $s^{12} = 1/729$) and $z = e^{i\pi/3}$, which lies on |s| = 1, does not shrink and cannot nest. The spirals are

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the steps s^t, $t \geq 1$, of the two turning wheels. (B) The four lanes of a layer agree on L_k leading hex digits (blue, slope 2.00 in $\log_{16} k$), and consecutive layers share a head (orange, slope 1.00); means of 300 random layers per point (Proposition 3).

4 Layers as atoms

In [1, §3–4] a layer is described from outside: four lanes nest, run, and deliver a value late. From inside, a layer is four copies of one motion. The lanes 1/(8k+1), 1/(8k+4), 1/(8k+5) and 1/(8k+6) ride the same circle under the same handle $x \mapsto \{16x\}$ and differ only in their denominators; the outer two differ by $5/((8k+1)(8k+6)) \approx 5/(64k^2)$, and the difference shrinks as k grows. Let L_k be the number of leading hex digits that all four expansions share, and write $\sigma_i = \Sigma_j\, j^i\, c_j$ for the moments of the weights.

Proposition 3 (copies that agree, then differ). *Let $k \geq 1$.*

(a) $H_k(m) = 0$ for $m < L_k$, and $H_k(L_k) > 0$. While the copies agree they emit identical crossings, which the weights cancel; the layer's signal starts at the first digit where they differ.

(b) $\lfloor \log_{16}(8k+1) \rfloor - 1 \leq L_k \leq \lfloor \log_{16}((8k+1)(8k+6)/5) \rfloor - 1$. The first part is the nest of [1, Prop. 3], the shared leading zeros; the rest is a lockstep run in which the four lanes cross together.

(c) $V_k = -\sigma_1/(8k)^2 + \sigma_2/(8k)^3 - \sigma_3/(8k)^4 + \ldots$, with $(\sigma_0, \ldots, \sigma_4) = (0, -15, -89, -465, -2429)$: the value of a layer is the first difference of the handle $1/(8k+y)$ across its copies.

Proof. (a) The (m+1)-th hex digit of 1/(8k+j) is the crossing $q_{k,j}(k+m)$. The four numbers decrease strictly with j, so at the first position where their digits are not all equal the digits satisfy $d_1 \geq d_4 \geq d_5 \geq d_6$ with $d_1 > d_6$, and there $H = 2(d_1 - d_4) + (d_1 - d_5) + (d_1 - d_6) > 0$; before it $H = (\Sigma_j\, c_j)\, d = 0$. (b) Leading zeros are shared, and numbers that share L leading digits differ by less than 16^{-L}. (c) Expand $1/(8k+j)$ in powers of $j/(8k)$. ■

Over 2,700 layers sampled on a logarithmic grid from $k \approx 320$ to $k \approx 2 \times 10^7$, the mean of L_k is $2.003 \log_{16} k - 0.06$, and H_k first becomes nonzero exactly at L_k in every case (Fig. 1B). Consecutive layers are copies too: V_k and V_{k+1}, aligned at their first nonzero digit, share a head of mean $0.998 \log_{16} k - 0.41$ digits, one more for every sixteenfold step in k, since $V_{k+1}/V_k = 1 - 2/k + O(k^{-2})$. This sharpens [1, Thm. 4]: until its copies split, a layer's signal is exactly zero, and the split and the value onset differ by O(1) digits.

This is the sense in which layers behave like atoms. Every layer is made from the same handle on the same ring, and its copies must differ, since identical copies cancel exactly ($\Sigma_j\, c_j = 0$). What a layer contributes is that difference, and it is heard only after the copies have run long enough to separate, about $2 \log_{16} k$ digits. The difference shrinks, so later layers are quieter and speak later, and successive layers repeat one another's leading digits. Nothing here needs more than one motion and a shrinking difference between its copies.

5 Complexity is nesting depth

Proposition 3(c) is one case of a general rule. For a formula $\Sigma_k\, b^{-k}\, \Sigma_j\, c_j/(Nk + j)$, let $\sigma_i = \Sigma_j\, j^i\, c_j$ and let ν be the number of leading moments that vanish: $\sigma_0 = \ldots = \sigma_{\nu-1} = 0 \neq \sigma_\nu$.

Proposition 4 (silent depth). *(a) $V_k = (-1)^\nu\, \sigma_\nu\, (Nk)^{-\nu-1}\, (1 + O(1/k))$, so layer k is silent in value for $(\nu + 1)\, \log_b(Nk) - \log_b|\sigma_\nu| + O(1)$ digits. Equivalently, the polynomial $\Sigma_j\, c_j\, y^{j-1}$, the numerator of the integrand of [1,*

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(8)] after the substitution $y = \sqrt{2}x$ (and its analogue in base b), has a zero of order ν at $y = 1$: the crest of [1, Thm. 5] is $\nu \geq 1$, and each further vanishing moment flattens it once more.

(b) Whatever the weights, the lanes of a layer agree on at most $\lfloor \log_b((Nk + j_{min})(Nk + j_{max})/(j_{max} - j_{min})) \rfloor - 1 = 2\log_b(Nk) + O(1)$ leading digits. So the value arrives with the first crossing when $\nu = 0$, at the split when $\nu = 1$, and about $(\nu - 1)\log_b(Nk)$ digits after the split when $\nu \geq 2$, the interval being filled by value-free carries.

(c) On the eight base-16 lanes $\nu \leq 1$ for every formula for π, with $\nu = 1$ only for BBP. On the twelve base-729 lanes $\nu \leq 2$ for $\pi\sqrt{3}$, and exactly one formula has $\nu = 2$:

$$9\pi\sqrt{3} = \Sigma_{k\geq 0}\, 729^{-k}\,(\,81/(12k+1) + 27/(12k+2) - 162/(12k+3) - 9/(12k+4) + 27/(12k+5)$$
$$+ 24/(12k+6) - 3/(12k+7) + 7/(12k+8) + 6/(12k+9) + 3/(12k+10) - 1/(12k+11)\,), \qquad (5)$$

with $\sigma_2 = 1092$ for these integer weights. In both systems the largest ν equals the number of independent relations.

Proof. (a) $1/(Nk + j) = \Sigma_i\, (-j)^i\, (Nk)^{-i-1}$. With $Q(y) = \Sigma_j\, c_j\, y^{j-1}$, σ_i is the value at $y = 1$ of $(y\, d/dy)^i\, (yQ(y))$, so $\sigma_0 = \ldots = \sigma_{\nu-1} = 0$ exactly when $yQ(y)$, and hence Q, vanishes to order ν at $y = 1$. (b) As in Proposition 3(b). (c) By Theorem 2 the binary formulas for π are $c(r) = b - rR$, with $\sigma_0 = -15r$ and $\sigma_1 = -15 - 67r$, which do not vanish together. On the ternary plane $\sigma_0 = 728/81 - 364b$ and $\sigma_1 = 1444/81 + 182a - 1632b$ vanish together only at $a = 10/81$, $b = 2/81$, which is (5) divided by 9. ■

Table 2. Anatomy of a layer. *Split:* mean position (0-based, in base-b digits) of the first digit at which the lanes of a layer differ, for $k \in [10^7, 2 \times 10^7)$. Then the mean of value onset minus split over 200 random layers $k \in [K, 2K)$, and the leading-order prediction $(\nu - 1)\log_b(1.5NK) + \log_b((j_{max} - j_{min})/|\sigma_\nu|)$ for $K = 10^7$.

Formula	ν	Split	$K = 10^3$	$K = 10^5$	$K = 10^7$	Predicted
BBP (1), base 16	1	11.95	−0.26	0.01	0.06	−0.40
30° wheel, base 729	0	4.96	−0.98	−2.23	−2.96	−2.87
Zero-sum (4), base 729	1	4.96	0.01	−0.27	0.04	−0.12
Double crest (5), base 729	2	4.93	1.02	1.78	2.27	2.51

So complexity, in the sense of how long a layer can hold its value before it must deliver it, is not an accumulation of terms but a count of cancellations. A layer can extend its nest only by preventing its value from being compiled early: each vanishing moment cancels one more order of what it would otherwise deliver at once. A layer with $\nu \geq 1$ nests for $\log_b(Nk)$ digits and runs in lockstep for about as many again; each vanishing moment beyond the first adds another $\log_b(Nk)$ digits of value-free carry; and the number of moments that can vanish is capped by the number of relations, which Theorem 2 counts by primes. BBP spends its single relation on exactly one cancellation. The ternary machine can afford two, and its double crest (5) keeps each layer silent for about $3\log_{729}(12k)$ digits, a third of them carry. In the picture of Section 1, the depth of the field a layer survives is set by structure, not by mass.

Remark (the traveler's gap is a nesting depth). The one-point recurrence of Bailey and Crandall [7] misreads digit n exactly when $\{16^n\pi\} < T_n$, with $T_n = \Sigma_{j\geq 0}\, V_{n+j}\, 16^{-j}$ [1, Thm. 16]; this needs the expansion of π after

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position n to open with $\lceil \log_{16}(1/T_n) \rceil - 1$ zeros. For $n \le 10^5$ this equals the silent depth $\lceil \log_{16}(1/V_n) \rceil - 1$ of layer n in 98.6% of cases and is one less in the rest: the traveler can misread only where π itself stays silent for as long as the next layer nests.

6 Chords and harmonics

A lane $d = 2^e o$ (o odd) runs on a coset aH of $H = \langle 16 \rangle$ in $(\mathbb{Z}/o)^\times$, and its loop mean μ leans by $\mu - \tfrac{1}{2} = (1/\phi(o)) \Sigma \chi(a^{-1}) S(\chi)$, summed over the odd Dirichlet characters χ with $\chi(16) = 1$ [1, Thm. 10]; balance, $\mu = \tfrac{1}{2}$, needs an involution in H [1, Thm. 12]. The characters are the harmonics of $(\mathbb{Z}/o)^\times$ that the motion leaves fixed. An involution $\varepsilon \in H$ pairs each point y of the loop with εy, the chords of [1, Fig. 3], and because $\chi(H) = 1$ forces $\chi(\varepsilon) = 1$, every chord restricts which harmonics can sound.

Proposition 5 (which harmonics a chord lets through). *In [1, Thm. 10]: (a) if H contains no involution, no odd character with $\chi(16) = 1$ is excluded, and their sum is not zero [1, Thm. 12b]; (b) if $-1 \in H$, every odd character is excluded, and $\mu = \tfrac{1}{2}$; (c) if H contains a partial inversion ε, with $\varepsilon \equiv 1 \pmod{o_1}$ and $\varepsilon \equiv -1 \pmod{o_2}$, the characters that contribute are exactly $\chi = \chi_1\chi_2$ with χ_1 odd modulo o_1, χ_2 even modulo o_2 and $\chi(16) = 1$: the surviving harmonics are odd on the fixed factor and even on the reflected one.*

Proof. $\chi(\varepsilon) = 1$ for every $\varepsilon \in H$. (b) $\chi(-1) = 1$ contradicts oddness. (c) o_1 and o_2 are coprime [1, Thm. 12d], so $\chi = \chi_1\chi_2$; then $\chi(\varepsilon) = \chi_2(-1) = 1$, and $\chi(-1) = \chi_1(-1)\chi_2(-1) = \chi_1(-1) = -1$. ■

This is [1, Thm. 12] heard rather than seen. A full chord silences every odd harmonic; a partial chord silences those that are odd on the reflected factor and leaves the fixed factor free to lean; with no chord nothing is silenced, and parity forbids what remains to sum to zero. We checked the description of the surviving characters directly on the lanes d = 204, 476, 493, 582 and 748. Which partial lanes the remaining harmonics balance is open [1, open problem 4], and one natural guess fails: balance is rarer, not commoner, when $16 \equiv 1$ on the fixed factor ($o_1 \mid 15$: 17 of 894 partial lanes balanced, against 576 of 8,817 otherwise, $k \le 20{,}000$).

Observation 6 (which loops carry the drift). *Split by loop type, the mean-field drift $M(1.28 \times 10^5) = 15{,}391.1$ of [1, (15)] comes 80.0% from lanes without involution (12,311.0), 20.1% from partial inversions (3,095.1), nothing from full inversions, and −15 from the two spent lanes $d = 1$ and $d = 4$.*

7 The same Mold in a compiler

7.1 Keeping the past without a stack

Is the Form particular to π? The test is the property the Mold was built for: a step that keeps its past in its present, with no stack and no side storage. On a finite state space a step does this exactly when it is a permutation, since then the present determines the past. The BBP lanes do it, because multiplication by 16 permutes $\mathbb{Z}/o$ for odd o [1, Thm. 9]. SHA-256 [18] does it too, by a Mold. Its state is eight 32-bit registers (a, …, h). Round t, with schedule word W_t and constant K_t, is

$$(a, b, c, d, e, f, g, h) \mapsto (T_1 + T_2, a, b, c, d + T_1, e, f, g), \tag{6}$$

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with $T_1 = h + \Sigma_1(e) + Ch(e, f, g) + K_t + W_t$ and $T_2 = \Sigma_0(a) + Maj(a, b, c)$, all sums mod 2^{32}. The 64 schedule words are compiled from the 16 words of the message block by

$$W_t = \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16} \quad (16 \le t < 64). \tag{7}$$

Theorem 7 (SHA-256 keeps its past without a stack). *(a) Each round is a Mold. The registers a, b, c, e, f, g are retained, shifted into b, c, d, f, g, h; the two oldest registers d and h are not stored but folded into the new ones, $e' = d + h + F_1$ and $a' = h + F_1 + F_2$, where $F_1 = \Sigma_1(e) + Ch(e, f, g) + K_t + W_t$ and $F_2 = \Sigma_0(a) + Maj(a, b, c)$ depend only on retained registers. Hence $h = a' - F_1 - F_2$ and $d = e' - h - F_1$, and every round is a permutation of the 2^{256} states.*

(b) The schedule is a Mold in the same sense: the step $(W_{t-16}, \ldots, W_{t-1}) \mapsto (W_{t-15}, \ldots, W_t)$ folds the oldest word into the new one, and $W_{t-16} = W_t - \sigma_1(W_{t-2}) - W_{t-7} - \sigma_0(W_{t-15})$. Any 16 consecutive schedule words determine the message block, and the block determines all 64.

(c) For every message block M the 64 rounds form a permutation E_M of the state, the block cipher SHACAL-2 [13]. The present state and the compiled program determine every earlier state.

Proof. (a) F_1 and F_2 are computed from registers that reappear unchanged in the new state; then h and d follow. (b) Solve (7) for W_{t-16}. (c) A composition of permutations. ■

Our implementation agrees with Python's hashlib on 1,003 messages; 1,000 random states and blocks were run forward through all 64 rounds and back to the start exactly, and 1,000 blocks were recovered from 16 consecutive schedule words at random offsets.

Where else. The general way to make a step keep its past out of a function that does not is the Mold, a shear: $(x, y) \mapsto (x \star F(y), y)$ is invertible whatever F is, because y is kept and F(y) can be computed again, and so is every composition of shears and swaps. Feistel networks are built this way [10]; so are Fredkin's second-order reversible automata [16, 20], the leapfrog integrators of Hamiltonian mechanics [12], the Hénon map $(x, y) \mapsto (1 - ax^2 + y, bx)$ for $b \neq 0$ [14], and Arnold's cat map, a product of two integer shears of the torus [4]. In the BBP machine the odd lanes are permutations outright, and π's phase becomes one only when the emitted digit is kept, which is the baker's map [1, Thm. 9]. What compilation adds is that the function F is itself compiled: in SHA-256 the shear of every round is keyed by a word of the program, and the program is compiled from the source by another shear.

7.2 Where distinctness is lost

If every round keeps its past, where does SHA-256 lose it? The compression of a block M with chaining value H is

$$f(H, M) = H + E_M(H), \tag{8}$$

the Davies–Meyer form [17, §9.4]: the permuted state is added back onto its own input.

Proposition 8 (one place of loss). *(a) In SHA-256, if $f(H, M) = f(H', M')$ with $(H, M) \neq (H', M')$, then either $M \neq M'$, and the output has dropped the program, or $M = M'$, $H \neq H'$ and $H + E_M(H) = H' + E_M(H')$, and the feed-forward has added back the input instead of keeping it. Nothing inside the 64 rounds can identify two states, since E_M is a permutation. (b) In the BBP machine every odd lane is a permutation and every even lane is one after its first step [1, Thm. 9a, b]; distinctness is lost only in the step $x \mapsto \{16x\}$ of π's phase, which drops the*

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emitted digit, the coordinate that the baker's map keeps [1, Thm. 9c]. In both machines the loss occurs at one place only, where the output drops the half of a Mold that was retained.

Proof. (a) $E_M(H) = E_M(H')$ implies $H = H'$. (b) is [1, Thm. 9]. ■

For a fixed chaining value, $M \mapsto f(H, M)$ maps 2^{512} blocks to 2^{256} values, so coincidences exist; by Proposition 8 they all arise at the output. That two inputs can share a value is what keeps arithmetic general, and SHA-256 admits it at that one place only: inside the rounds the past is kept separate absolutely. The same holds for π. The lift $16^n\pi = \lfloor 16^n\pi \rfloor + \{16^n\pi\}$ holds the history in its integer part and the now in its fraction; multiplication by 16 is invertible on the lift, and only the projection that keeps the now alone makes the step 16-to-1. The one-way step is not a defect of either machine; it is where the output leaves.

7.3 The field of a message word

Now the runner. Let two message blocks differ in word j only, $0 \le j \le 15$, by $\delta = W'_j - W_j \neq 0 \pmod{2^{32}}$, and write depth(r) = 0, 1, 2, 3 for the register pairs (a, e), (b, f), (c, g), (d, h).

Proposition 9 (first contact, delay line, survival). *(a) The two state sequences agree before round j. After round j exactly a and e differ, both by the additive amount δ.*

(b) Register r after round t depends on W_j, as a formula, if and only if $t \ge j + \text{depth}(r)$: the change crosses the delay line, from a to d and from e to h, in three rounds and is folded back into the new registers at round j + 4, through h in T_1 and d in e'.

(c) The two state sequences remain distinct at least through round $\tau_j - 1$, where τ_j is the first round whose schedule word depends on W_j: $\tau_j = 16$ for j = 0, 1, 9, 14, $\tau_j = j + 15$ for $2 \le j \le 8$, $\tau_j = j + 7$ for $10 \le j \le 13$, and $\tau_{15} = 17$.

Proof. (a) Before round j nothing differs; in round j only W_j does, so T_1 changes by δ and T_2 and d do not. (b) The shift moves a and e one register per round, and from round j + 1 on the new a and e are formulas in registers that already depend on W_j. (c) In rounds j + 1, …, $\tau_j - 1$ both runs apply the same permutation (Theorem 7a), which maps distinct states to distinct states. The values of τ_j follow from (7). ■

After τ_j the two runs execute different programs, and a later schedule difference can in principle cancel a state difference; this is the local-collision mechanism of differential analysis [11]. Within the window the runner cannot be removed from the field.

Observation 10 (closure, survival and the value front). *(a) Structurally, every one of the 256 state bits depends on every bit of W_j after round j + 5, for every j, and on every bit of the chaining input after round 8; this is computed from the code alone, with each addition making bit k depend on bits 0, …, k of its addends. (b) In values, over 10,752 one-bit changes (each of the 512 message bits, on the zero block and on 20 random blocks), Proposition 9(a) held in every case; the state difference never vanished at any round; all eight registers differed exactly three rounds after first contact in every case; the mean number of differing state bits r rounds after first contact was 3.9, 25.0, 55.7, 87.6, 115.7, 126.6 and 128.0 for r = 0, …, 6 and stayed between 127.6 and 128.2 afterwards; and the digests differed in 127.97 of 256 bits on average. (c) The value front is not forced by the structure. Placing a 19-bit carry run in e and choosing f and g so that Ch cancels the change of Σ_1, we built a block in which flipping bit 0 of W_6 leaves register g equal after round 9 = j + 3, although g depends on W_6 from then on; the state difference survives, and the digests differ in 136 bits.*

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Such a lag needs a long carry. It is impossible when the change does not carry inside e, because Σ_1 moves the changed bit to three other positions and the lowest of four distinct signed powers of 2 cannot cancel; the shortest carry run for which our search found one is 14 bits. Figure 2 shows the field of every word. The starting point fixes how much field lies ahead, 64 – j rounds. The field fixes what is crossed: the four-stage delay line, then closure five rounds after contact, the same for every word. And closure arrives on its own. A one-bit change, the smallest runner there is, survives the whole field and arrives as a change of about half of the 256 output bits. One change of the source reaching every bit of the destination, while distinct sources stay distinct inside the rounds, is what we mean by entangled but separate.

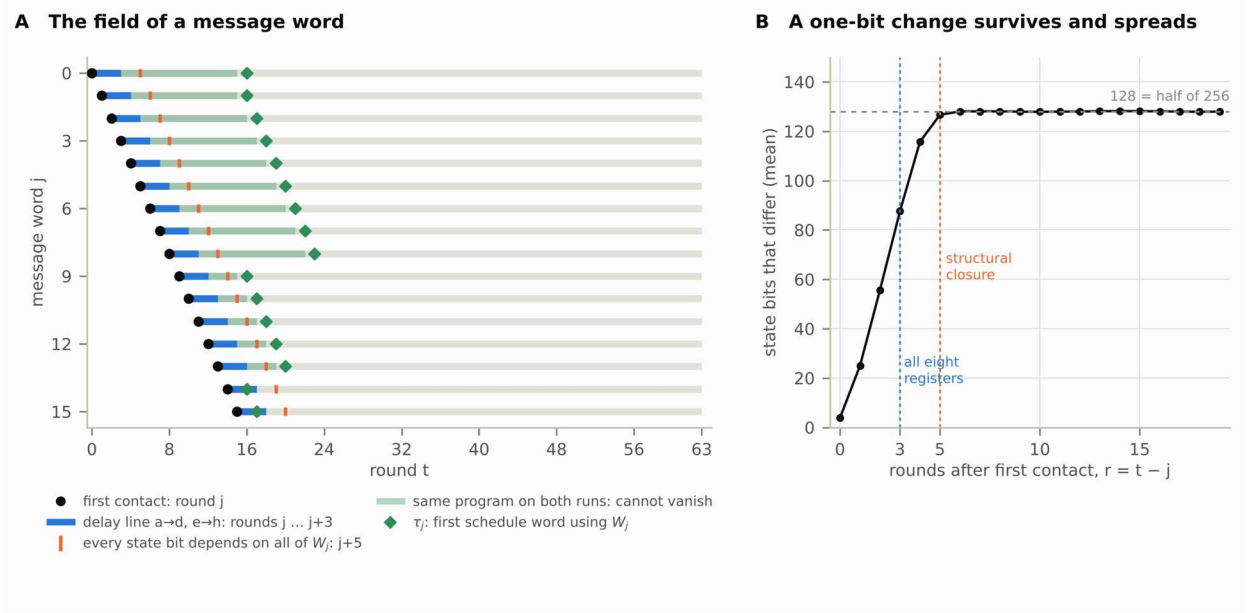


Figure 2. The field of a message word in SHA-256. (A) For each word j: first contact at round j (dot); the delay line from a to d and from e to h (blue, rounds j to j + 3); structural closure (orange tick, round j + 5: every state bit depends on every bit of W_j); the window in which both runs execute the same program, so that a difference cannot vanish (green); and τ_j, the first schedule word that uses W_j (diamond). (B) Mean number of differing state bits r rounds after first contact, over 10,752 one-bit changes; the dashed lines mark the structural front of all eight registers (r = 3) and closure (r = 5).

7.4 One Form, two machines

Table 3. The same Form in the BBP machine [1] and in SHA-256.

Form	BBP machine	SHA-256
Compile first	Every lane present at step 0 [1, Prop. 1]	The block is compiled into 64 schedule words by a Mold (Thm. 7b)
Nest	Lane (k, j) silent until step k + $\lceil \log_{16} d \rceil - 1$ [1, Prop. 3]	Word j silent until round j (Prop. 9a)
Run	First crossing	Round j: a and e change by the same amount
Lockstep, delay	Four copies agree on about 2 $\log_{16}$ k digits (Prop. 3)	The change crosses from a to d and from e to h in three rounds (Prop. 9b)

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Form	BBP machine	SHA-256
Closure	Value arrives near the split (Prop. 4)	Every state bit bound to every bit of W_j at round $j + 5$ (Obs. 10)
Retain (Mold)	new = 16·old – emitted; lanes are permutations [1, Thm. 9]	Rounds and schedule are Molds; 64 rounds are a permutation (Thm. 7)
Survive	Silent depth $(\nu + 1)\log_{16}(8k)$ (Prop. 4)	A difference cannot vanish before round τ_j (Prop. 9c)
Loss	$x \mapsto \{16x\}$ drops the emitted digit (Prop. 8b)	The output drops the program and adds back the input (Prop. 8a)
Read	Digit = difference of crossings [1, Prop. 2]	Digest = state + input (8)

The two machines differ in where the depth lives. In BBP the field deepens with the starting point: the silent depth of layer k grows like $(\nu + 1)\log_{16}(8k)$. In SHA-256 the field is uniform: three rounds of delay line and five to closure for every word, and the starting point fixes only how much field remains and when the program injects the change again.

8 Discussion

The question behind both papers is whether the Form is a property of π or of transformation, and the answers here point to the second. A rational digit machine can turn only by crystallographic angles, and the quarter turn is the one that nests in a power of 2. Its lanes have exactly one way to trade against each other, and BBP spends it on one cancellation. A layer is four copies of one motion, silent while they agree and heard through their difference, and a loop leans unless a chord silences its odd harmonics. A compiler shows the same Form for its own reasons: SHA-256 is built from Molds at both scopes, keeps its past without a stack, and loses distinctness only where its output drops what the Mold retained, the same place where π's phase loses it. A complex process, in both machines, is not accumulated mass but one that survives its field, and how far it can survive is set by structure: by cancellation in BBP, and by the delay line and the program in SHA-256.

Open problems

1. For which fold systems besides the two of Theorem 2 do the lanes span exactly as many dimensions as there are logarithms behind the fold points, and is the largest number of vanishing moments always the number of relations? Rotations outside the crystallographic set lead to irrational bases [15]; do their lane systems obey the same count?
2. Which partial-inversion cosets balance [1, open problem 4]? Balance is rarer when $16 \equiv 1$ on the fixed factor (Section 6); is there a density?
3. The asymptotic law of the drift of the carry [1, open problem 3], and why lanes without involution carry four fifths of it (Observation 6).

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4. How far can the value front of SHA-256 lag its structural front for a one-bit change (Observation 10c), and what is the shortest carry run that allows a lag?
5. The Bailey–Crandall conjecture and the normality of π in base 16 [1, 7]. By the remark of Section 5, the traveler can misread only where π's own expansion is silent for as long as the next layer nests.

Where the Form comes from, whether a first difference compels inversion, is still outside what either machine can show. What both show is that wherever an inversion appears it is the half-period of an even closure [1, §8], and wherever the past is kept it is kept by a Mold.

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Appendix: Reproducibility

All computations use Python 3 with mpmath, sympy and exact rational arithmetic, plus one C program. The scripts and their logs accompany this paper as survive-field-scripts.zip; times are for one CPU core.

Table 4. Scripts.

Script	Verifies	Time
why/fold_wheels.py	Thm. 1: the 45° and 30° wheels, their formulas, turn sums and seams	1 s
why/rigidity.py	Thm. 2: closed forms of all lanes, branch checks, exact rank and null space	1 s
why/relations.py, ternary*.py	Thm. 2 cross-checks by PSLQ; the $\pi\sqrt{3}$ plane, (4) and (5)	35 s
why/atoms_grid.py	Prop. 3: L_k, signal onset, shared heads (Fig. 1B); the remark of Section 5	5 s
why/anatomy.py	Prop. 4: Table 2	2 s
why/partial_structure.py	Prop. 5 on five lanes; balance by $o_1 \mid 15$, $k \le 20{,}000$	80 s
why/drift_types.c	Obs. 6: the drift split by loop type, $k \le 128{,}000$	80 s
why/fig_why.py	Figure 1	1 s
sha/sha_mold.py	Thm. 7: against hashlib; rounds and schedule run backwards	1 s
sha/sha_front.py	Prop. 9(b, c), Obs. 10(a): the location tensor from the code	1 s
sha/sha_survival.py	Prop. 9(a), Obs. 10(b): 10,752 one-bit changes	7 s
sha/front_lag.py	Obs. 10(c): the block in which the value front lags; shortest carry run	4 s
sha/fig_sha.py	Figure 2	2 s

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